HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Department of Actuarial Mathematics and Statistics

F71BF

Life Insurance Mathematics 2

Semester 2 2017/18

Duration: Two Hours

You should answer ALL SEVEN questions.

Total 60 marks.

You should be provided with the yellow Formulae and Tables for Actuarial Examinations.

You may use a permitted calculator.

Question 1

A 20-year endowment is sold to a man age 45, with sum assured £100,000 paid at the end of the year of death or on maturity, and with level annual premiums payable in advance.

(a) Calculate V(13.7), the policy value at time 13.7. You should state any assumptions that you use.

Basis:
Method: Net premium valuation
Mortality: AM92 ultimate
Interest: 4% per year effective [7 marks]

(b) Give two reasons why a life office might need to calculate a policy value at a time that is not an exact number of years. [2 marks]

[Total 9 marks]

Question 2

The following table gives a summary of mortality for workers in a company, and for the country as a whole.

	Company		Whole country	
Age group	Population	Deaths	Population	Deaths
18-35	1,600	3	50 million	105,000
36-55	1,400	8	42 million	330,000
56-70	2,400	64	33 million	950,000
Total	5,400	75	125 million	1,385,000

(a) Calculate the crude death rate for the company. [1 mark]

(b) Calculate the crude death rate for the whole country. [1 mark]

(c) Calculate the standardised mortality ratio for the company, using the whole country as the standard population. [3 marks]

(d) Explain why the standardised mortality ratio might be preferred to crude death rates. [1 mark]

[Total 6 marks]

Question 3

Describe the following common characteristics of income protection insurance, and give reasons for their use in policy design:

(a) deferred period [2 marks]

(b) off-period [2 marks]

[Total 4 marks]

Question 4

A model for an individual long-term care insurance policy has the following four states:

State 0: Independent Living
State 1: Assisted Living in their own home
State 2: Care Home
State 3: Dead

The model uses the following transition intensities, which are all constant:

$\mu^{01} = 0.02$
$\mu^{03} = 0.03$
$\mu^{12} = 0.20$
$\mu^{13} = 0.30$
$\mu^{23} = 0.60$

All other transition intensities are zero.

(a) Draw a state diagram for this model. [2 marks]

(b) Calculate the probability that an individual who is now living independently will, in 10 years' time, require assistance to live at home. [6 marks]

[Total 8 marks]

Question 5

A joint whole-life assurance policy provides a benefit of £1 immediately on the death of the last survivor of a male, now aged 60 exact, and a female, now aged 57 exact.

Calculate:

(a) The expected present value of this policy. [4 marks]

(b) The standard deviation of the present value of this policy. [4 marks]

Basis:

Forces of mortality: male life- a constant force of 0.04
female life- a constant force of 0.03

Force of interest: constant, 0.05

[Total 8 marks]

Question 6

A life office issues a large number of 4-year unit-linked endowment assurance policies, to lives aged 60 exact. Level premiums are paid annually in advance, until maturity or earlier death.

The office has performed a profit test on these policies and the profit vector per policy sold, ignoring surrenders, is as follows:

(200, -120, -10, 30)

The following basis is used for all parts of this question:

Mortality	$q_x = 0.01$ at all ages
Interest on sterling cashflows	7% per year
Expenses	None

(a) Calculate the profit signature per policy sold, if negative sterling fund cash flows are zeroised. [3 marks]

The company now wishes to allow for surrenders in its calculations. It assumes that at the end of the first year only, 10% of the surviving policyholders will surrender. Surrender values are equal to the bid value of units held (after deduction of the fund management charge) less a surrender penalty of 70.

(b) Calculate the revised profit signature per policy sold after allowing for surrenders, if negative sterling fund cash flows are zeroised. [6 marks]

(c) Calculate the net present value of the profit signatures in parts (a) and (b), using a risk discount rate of 12% per year. [2 marks]

(d) Calculate the difference between the two net present values in (c). Give two reasons why allowing for surrenders creates this difference. [3 marks]

[Total 14 marks]

Question 7

[Subsequent note: from September 2018 the syllabus is changing. It is likely that this question will no longer be on the syllabus.]

A pension fund gives an age retirement pension of 1/90th of final pensionable salary for each year of service. Final pensionable salary is defined as the average of the salaries earned in the last three years before retirement. All 500 members of the fund are age 43, with exactly 8 years' past service, and a current salary rate of $45,000 per year. You should use the service table in the Formulae and Tables for Actuarial Examinations, and assume that that salaries are revised continuously.

(a) Calculate the total expected present value of members' past service age retirement pensions. [3 marks]

(b) Calculate the total expected present value of members' future service age retirement pensions . [3 marks]

(c) If the scheme's assets total $9,000,000, calculate the contribution rate k% as a percentage of members' salaries which is required to fund members' total retirement benefits by the time they retire. [5 marks]

[Total 11 marks]

END OF PAPER